

ST Project:

Build an Automation Framework using Playwright with JS

Members:

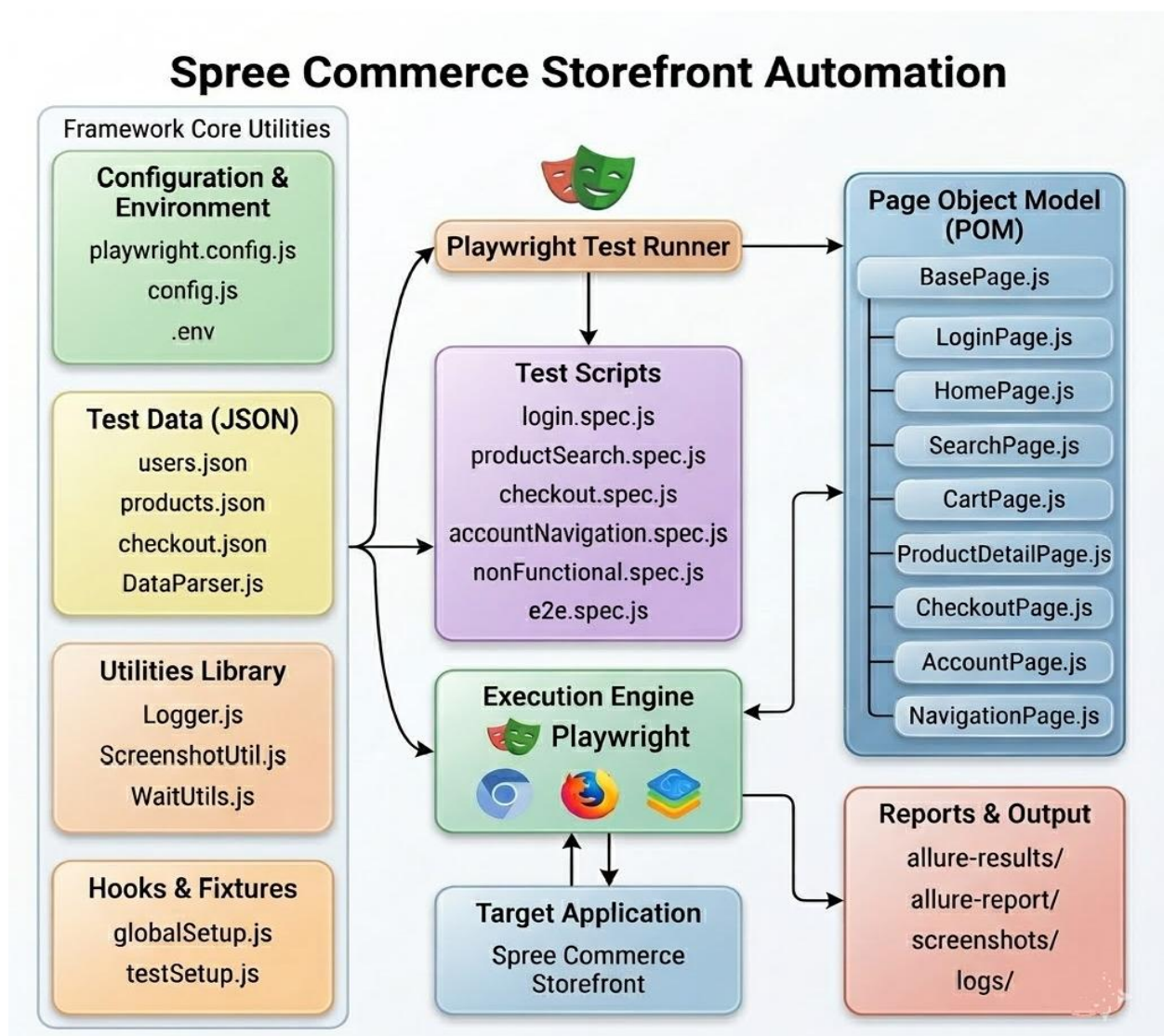
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1. Automation Framework Diagram:



2. File Hierarchy:

```
'''directory
spree-playwright-framework/
├── .env                # Key environment variables (BASE_URL, email, password)
├── package.json        # Project dependencies, scripts, and libraries
├── playwright.config.js # Core Playwright configuration (timeouts, reports, devices)
├── framework_diagram.png # Framework architecture flow diagram image
├──
├── config/
│   └── config.js      # Configuration manager parsing env variables
├──
├── data/              # JSON structured test datasets (Data-Driven Testing)
│   ├── users.json    # Positive and negative credentials
│   ├── products.json # Search keywords and cart inputs
│   └── checkout.json  # Shipping addresses, promo codes, and cards
├──
├── fixtures/
│   └── testSetup.js   # Playwright custom page object fixtures injection
├──
├── hooks/
│   └── globalSetup.js # Pre-run global setup validation
├──
├── pages/             # Page Object Model (POM) Layer
│   ├── BasePage.js   # Safe low-level wrapper methods (click, fill, wait)
│   ├── HomePage.js   # Home layout headers, search boxes, and navigation
│   ├── LoginPage.js   # Authentication form actions
│   ├── SearchPage.js  # Results grids, filters, and sorters
│   ├── ProductDetailPage.js # Add to cart buttons, variants, and quantities
│   ├── CartPage.js    # Shopping cart drawer and checkout redirection
│   ├── CheckoutPage.js # Shipping address, delivery methods, and payment steps
│   ├── AccountPage.js # User dashboard navigation and edit forms
│   └── NavigationPage.js # Global header/footer layout linkages
├──
├── tests/             # Test Scripts Layer (43 Total Test Cases)
│   ├── login.spec.js  # TC-01 to TC-10 (Authentication scenarios)
│   ├── productSearch.spec.js # TC-11 to TC-20 (Product filtering & cart counts)
│   ├── checkout.spec.js # TC-21 to TC-30 (Order placement validation)
│   ├── accountNavigation.spec.js # TC-31 to TC-40 (User dashboard navigation checks)
│   ├── nonFunctional.spec.js # TC-41 to TC-42 (Viewports responsiveness checks)
│   └── e2e.spec.js    # TC-43 (End-to-End customer journey checkout flow)
├──
├── utils/             # Reusable Utilities Layer
│   ├── DataParser.js  # JSON parser helper
│   ├── Logger.js      # Winston logger configuration
│   ├── ScreenshotUtil.js # Screenshot capture and Allure report attacher
│   └── WaitUtils.js    # Custom dynamic explicit wait conditions
├──
├── logs/              # System text logs generated during execution
'''
```

3. Playwright test execution:

All test cases passed in 3 browsers

```
Slow test file: [chromium] > tests\checkout.spec.js (5.1m)
Consider running tests from slow files in parallel. See: https://playwright.dev/
2 flaky
[firefox] > tests\checkout.spec.js:83:3 > Checkout + Order Flow Tests - Spree Commerce > TC-23: Verify checkout fails if invalid postal code is entered
[firefox] > tests\nonFunctional.spec.js:16:3 > Non-Functional Tests - Spree Commerce > TC-20: Verify checkout fails if invalid postal code is entered
127 passed (16.6m)
```

4. Playwright report:

▼ checkout.spec.js		
⚠️	Checkout + Order Flow Tests - Spree Commerce > TC-23: Verify checkout fails if invalid postal code is entered (firefox)	1.2m
checkout.spec.js:83 [▶] [≡] View Trace		
✓	Checkout + Order Flow Tests - Spree Commerce > TC-21: Verify user can proceed to checkout with valid cart items (chromium)	33.8s
checkout.spec.js:29		
✓	Checkout + Order Flow Tests - Spree Commerce > TC-22: Verify checkout fails if required shipping fields are empty (chromium)	32.1s
checkout.spec.js:54		
✓	Checkout + Order Flow Tests - Spree Commerce > TC-23: Verify checkout fails if invalid postal code is entered (chromium)	34.9s
checkout.spec.js:83		
✓	Checkout + Order Flow Tests - Spree Commerce > TC-24: Verify user can select shipping method successfully (chromium)	33.2s
checkout.spec.js:112		
✓	Checkout + Order Flow Tests - Spree Commerce > TC-25: Verify user can apply valid discount/promo code (chromium)	33.1s
checkout.spec.js:142		
✓	Checkout + Order Flow Tests - Spree Commerce > TC-26: Verify invalid promo code is rejected (chromium)	33.2s
checkout.spec.js:169		
✓	Checkout + Order Flow Tests - Spree Commerce > TC-27: Verify order confirmation page appears after successful (chromium)	31.8s

5. Allure Report:

Overview

Categories

Suites

Graphs

Timeline

Behaviors

Packages

ALLURE REPORT

6/15/2026

16:44:16 - 16:56:36 (12m 19s)

129

test cases

100%

SUITES 3 items total

chromium

43

firefox

43

webkit

43

Show all

TREND

There is nothing to show

CATEGORIES 0 items total

Show all

Overview

Categories

Suites

Graphs

Timeline

Behaviors

Packages

Suites

order name duration status

Status: 0 0 129 0 0 Marks: [icons]

chromium 43

accountNavigation.spec.js 10

Account + Navigation + UI Tests - Spree Commerce 10

#1 TC-31: Verify user can view profile/account page chromium... 16s 603ms

#2 TC-32: Verify user can navigate to profile section chromium... 11s 205ms

#3 TC-33: Verify user can navigate to order history chromium... 10s 740ms

#4 TC-34: Verify user can navigate to addresses page chromium... 23s 620ms

#5 TC-35: Verify navigation between Home → Products → Cart works correctly chromium... 12s 151ms

#6 TC-36: Verify page reload maintains correct route state chromium... 9s 927ms

#7 TC-37: Verify broken/invalid URL shows 404 or error page chromium... 4s 165ms

#8 TC-38: Verify homepage UI loads correctly with header and footer chromium... 13s 212ms

#9 TC-39: Verify unauthorized access to order history is blocked chromium... 5s 158ms

Execution

Test body

Before Hooks 8 sub-steps 1s 935ms

Step 1: Login with valid credentials 12 sub-steps, 1 attachment 15s 764ms

Step 2: Verify account page loads with user info 4 sub-steps, 1 attachment 136ms

Wait for selector locator("button:has-text('Sign Out')").first() 8ms

Expect "toBeTruthy" 0s

Screenshot 103ms

TC_31_step2_account_page 90.8 KiB

Account Overview

Order History

Addresses

Payment Methods

Profile